Topic: Word problems into equations

Question: Bob's age is three less than twice Anne's age. If Anne is 15, How old is Bob?

Answer choices:

A 9

B 27

C 30

D 12

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Solution: B

In this problem, we have two quantities: Anne's age and Bob's age, so let's let A be Anne's age, and let B be Bob's age.

Now we need to translate each word or phrase into mathematical symbols. Here, "Bob's age is three less than twice Anne's age."We can translate this as "$B = 2A - 3$."

We're given Anne's age as 15, so we substitute 15 for A and then simplify to get B.

$$B = 2(15) - 3$$

$$B = 30 - 3$$

$$B = 27$$

Bob is 27.

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Topic: Word problems into equations

Question: Find the value of the expression.

$\frac{3}{4}$ of 200

Answer choices:

A 170

B 200

C 150

D $\frac{4}{3}$

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Solution: C

When we're turning a phrase into a mathematical expression or equation, the word "of" immediately after a fraction tells us to multiply. Therefore,

$$\frac{3}{4}(200)$$

$$\frac{3(200)}{4}$$

$$\frac{600}{4}$$

$$150$$

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Topic: Word problems into equations

Question: Eight less than four times a number is 92. Find the number.

Answer choices:

A -25

B 20

C 100

D 25

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Solution: D

When we're turning a phrase into a mathematical expression or equation, the word "times" tells us to multiply. The phrase "less than" tells us to subtract, and the word "is" means equals. Therefore, the equation will be

$$4x - 8 = 92$$

Use inverse operations to solve for the number.

$$4x - 8 + 8 = 92 + 8$$

$$4x = 100$$

$$\frac{4x}{4} = \frac{100}{4}$$

$$x = 25$$

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